

Experiment # Title

AUTHOR

Date Performed:

Partners:

Date Submitted:

1 Objective of the experiment

Provide a clear, concise description of the experiment.

Example:

- (i) To verify Kirchhoff's Voltage Law (KVL) and Kirchhoff's Current Law (KCL) on a given network.
- (ii) To obtain Thevenin's equivalent circuit of network given in Fig. 1 at the load terminals (C-D).

2 Apparatus used in the experimental setup

List the instruments used in the experiment.

List the components used in the experimental setup.

If applicable, mention the software used for simulation.

3 Circuit diagram along with proper labeling

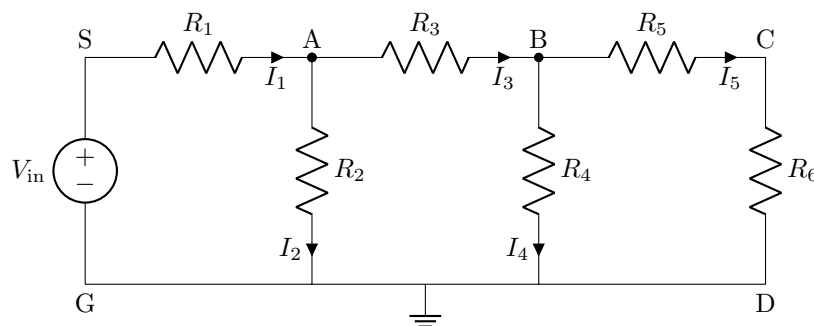


Figure 1: Circuit diagram for experiment 1 (made using circuitikz in Latex)

4 Theoretical formulations

Explanation of the theoretical concepts and derivations involved.

Mathematical formulas for the relevant concepts.

Example. Kirchhoff's Current Law (KCL):

At any instant, the algebraic sum of currents into a node is zero. It indicates the “Law of conservation of charge”. Mathematically,

$$\sum_{k=1}^n I_k = 0, \quad (1)$$

where n is the total number of branches connecting at a node and I_k is the current flowing into the node from branch k .

5 Measurements/Experimental data

List/tabulate your observations with proper units

Table 1: Resistances

Resistance	R_1	R_2	R_3	R_4	R_5	R_6
Ideal (Ω)						
Measured (Ω)						
Tolerance						

Table 2: Observations: For KCL verification (referring to Equation (1))

Source Voltage (V_{in})	Node A				Node B			
	I_1	I_2	I_3	$I_1 - (I_2 + I_3)$	I_3	I_4	I_5	$I_3 - (I_4 + I_5)$

Table 3: Parameters for Thevenin's equivalent circuit (errors can be computed in Latex with spreadtab)

	Quantity	Measured value	Calculated value	Error (= Measured - Calculated)	Percentage Error (= $\frac{\text{Error}}{\text{Calculated}} \times 100$)
1	Thevenin voltage				
2	Thevenin resistance				
3	Short-circuit current				

6 Sample calculations and verifications

A few sample calculations

Example 1. Calculation of currents from voltage measurements

Example 2. Theoretical values of voltages and currents

7 Concluding remarks

- Is the objective of the experiment accomplished?
- List any additional precautions taken during the experiment.
- Identify possible sources of error.
- Discuss the practical applications of the experiment.
- Discuss the future scope of this experiment or related experiments.

References

List any sources or references used in preparing the lab report. Include textbooks, journal articles, websites, or any other relevant resources.